

AI Voice Agent

Project Journey & Engineering Case Study

FROM IDEA → LIVE CALL

A complete account of the problem, product decisions, voice-agent design, architecture, telephony integration, structured lead capture, dashboard, debugging, validation, and final outcome.

The product in one sentence

Turn a submitted phone number into an immediate, human-feeling AI conversation — then turn that conversation into an actionable Sunrise Interiors lead.

Dimension	Final implementation
Conversation	60–90 second outbound qualification call
Languages	English, Hindi and Hinglish
Voice AI	Sarvam Voice Agents — Shubh
Telephony	Exotel
Frontend	Next.js 16.3.4 + Tailwind
Backend	FastAPI
Operations	Lead dashboard + call lifecycle

The Starting Point

The assignment was simple in wording but demanding in execution.

Brief: Sunrise Interiors needs to respond quickly after a visitor shares a phone number for interior-design interest. The system should call immediately, introduce itself, check whether it is a good time, understand the work needed, ask when the customer wants to start, offer a designer meeting, and close naturally.

The actual engineering problem

The challenge was not merely placing a phone call. It was creating a low-latency, multilingual, interruption-tolerant conversation and connecting it reliably to lead persistence and an operator-facing outcome.

Requirements → decisions

Requirement	Decision
Immediate callback	Form submission triggers outbound Exotel call.
Human feel	Natural Indian voice, short turns, acknowledgements, interruptions.
60–90 seconds	Only high-value qualification questions.
English / Hindi / Hinglish	Auto-detection + quick switching.
Bonus lead extraction	Structured project, timeline, language and meeting fields.
Meeting confirmation	Explicit meeting_requested value.
Operational visibility	Dashboard backed by the same backend.

Conversation Design

The call was deliberately designed as a conversation, not a questionnaire.

The agent persona is **Shubh**, a warm professional AI design concierge calling from Sunrise Interiors.

“Hi, this is Shubh calling from Sunrise Interiors. Is this a good time for a quick minute?”

1 Check whether the customer can speak.
2 Understand the interior work required.
3 Ask when the customer wants to start.
4 Offer a Sunrise Interiors designer consultation.
5 If accepted, acknowledge and say a designer will follow up.
6 If declined, thank the customer and close politely.

Design choice	Why
One question at a time	Avoids questionnaire behaviour.
Interruptions enabled	Allows natural turn-taking.
Brief acknowledgements	Adds human rhythm without bloating duration.
Language switching	Matches natural Indian English/Hindi/Hinglish use.
Explicit closing rule	Prevents an abrupt or robotic ending.
Bounded duration	Prevents runaway calls.

Choosing the Technology

Each component was selected for a specific part of the experience.

Layer	Technology	Purpose
Frontend	Next.js 16.3.4	Phone capture, call flow and dashboard.
Backend	FastAPI	Lead API, call trigger, webhooks and dashboard APIs.
Voice	Sarvam Voice Agents	Realtime Indian voice, multilingual conversation, tools.
Telephony	Exotel	Outbound calls, caller ID, VoiceBot routing, callbacks.
Public bridge	ngrok	Expose the local API to external voice/telephony services.

Why Sarvam?

The assignment required natural English, Hindi and Hinglish. Sarvam provided a practical realtime voice-agent environment with Indian-language handling, interruptions, tools and telephony integration, avoiding the need to assemble a complete custom realtime STT → LLM → TTS pipeline for a short assignment.

Why Exotel?

The experience had to reach a real phone number. Exotel supplied the telephony boundary and VoiceBot routing required for a real outbound call.

Why FastAPI?

The backend needed a small, explicit, testable API surface rather than a large application framework.

End-to-End Architecture

The system is a chain of independently testable boundaries.

1	Visitor opens Sunrise Interiors page.
2	Visitor submits phone number.
3	Next.js → POST /api/leads.
4	FastAPI creates lead and requests Exotel outbound call.
5	Exotel starts the configured flow and connects to Sarvam VoiceBot.
6	Sarvam runs the Shubh conversation.
7	Sarvam invokes save_lead_details with extracted fields.
8	FastAPI /api/voice/save-lead updates the lead.
9	Exotel status callback updates call lifecycle.
10	Dashboard reads and displays the resulting lead.

Component	Responsibility
Next.js	Customer interaction and presentation.
FastAPI	Business API, persistence, call initiation, webhooks.
Exotel	Phone network and call lifecycle.
Sarvam	Realtime conversational intelligence and extraction.
Dashboard	Operator visibility and lead outcome.

Core insight

The AI voice model is only one component. The product is the complete loop: intent → call → conversation → structured data → persistence → operator action.

Voice-Agent Engineering

The final configuration prioritizes naturalness and short-call efficiency.

Setting	Configuration
Persona / voice	Shubh
Speed	1.10×
Languages	Hindi + English
Mode	Auto-detect + quick switching
Start	English
Interruptions	ON
Eagerness	Balanced
Volume	Normal
Background	Quiet office
Indic numbers	ON
Maximum call	3 minutes
Unanswered nudges	Disabled

Prompt strategy

The instructions emphasize one question at a time, concise responses, natural Indian English, English/Hindi/Hinglish switching, interruption handling, and no IVR/scripted behaviour. The closing is explicitly controlled: once meeting intent is established, the agent does not ask another qualification question.

Why the final seconds matter

A technically correct voice flow can still feel robotic. The closing was therefore designed explicitly so the agent acknowledges the customer's answer and gives the conversation a natural endpoint.

Structured Lead Extraction

The voice conversation becomes business value only when useful information survives the call.

Variable	Meaning
project_type	Latest, most specific interior requirement; multiple areas preserved.
timeline	Latest answer about when the customer wants to start.
preferred_language	English, Hindi or Hinglish based on actual usage.
meeting_requested	True only for explicit agreement to a designer consultation.

Tool: save_lead_details

The Sarvam tool POSTs structured fields to `/api/voice/save-lead`. The backend maps them to the existing lead and updates qualification/call state.

1 Conversation answer
2 Agent variable
3 save_lead_details
4 POST <code>/api/voice/save-lead</code>
5 Lead update
6 Dashboard outcome

Verified live result

A successful call produced: living room and kitchen renovation; two-three months later; Hinglish; meeting_requested=true. The webhook returned 200 OK and the dashboard showed the lead as Qualified with a designer meeting requested.

Telephony Integration

The hardest part was making every boundary work together.

Element	Role
Exotel caller ID	Presents the configured Sunrise calling number.
Call Connect API	FastAPI requests the outbound call.
Exotel flow	Routes the call into the VoiceBot.
Sarvam VoiceBot	Streams the live conversation.
Status callback	Sends call lifecycle state to FastAPI.
Recording	Completed recorded calls can expose a recording URL.

Call lifecycle

1 POST /api/leads → lead created
2 trigger_outbound_call() → Exotel
3 Exotel → Sarvam VoiceBot
4 Caller ↔ Shubh
5 Sarvam → save_lead_details
6 FastAPI → lead updated
7 Exotel → status callback
8 Dashboard → operator view

The Debugging Journey

The final reliability came from isolating failures instead of changing the whole stack.

Public webhook 404: local FastAPI worked, but the public ngrok path initially returned Not Found.

ngrok browser warning: a direct request produced ERR_NGROK_6024. Testing with the skip-browser-warning header showed that the warning was separate from the application.

Port/process discovery: Windows showed three listeners associated with port 8000: Docker Desktop, WSL relay, and the Python Sunrise API. The API itself was listening on 127.0.0.1:8000.

Resolution: ngrok was explicitly pointed at the intended listener:

```
ngrok http 127.0.0.1:8000
```

Final proof

After the tunnel was corrected, the public /api/voice/save-lead endpoint returned HTTP 200 with success=true. This proved the public webhook could reach the actual Sunrise FastAPI service.

Lesson: test local API → public tunnel → webhook → telephony → voice agent → persistence as separate boundaries.

Validation & Evidence

Validation was performed across both individual endpoints and the real phone path.

Check	Observed result
Local /health	200 OK
Public /health	200 OK through ngrok
Public /api/voice/save-lead	200 OK, success=true
POST /api/leads	201 Created
Exotel → Sarvam	Connected to VoiceBot
Sarvam extraction	Payload reached backend on successful call
Exotel status	completed callback observed
Dashboard	Structured lead displayed

Representative successful call log chain

1 POST /api/leads 201 Created
2 SARVAM SAVE-LEAD PAYLOAD received
3 POST /api/voice/save-lead 200 OK
4 EXOTEL STATUS: completed
5 Recording URL returned
6 Dashboard updated

Important correctness fix

The persistence layer was updated to correctly interpret string boolean values such as “true” for meeting_requested, preventing valid Sarvam output from being treated as false.

Earlier tests also exposed no-answer and very short calls. Those outcomes are tracked separately from qualification, so a completed phone connection is not automatically treated as a qualified lead.

Dashboard & Business Outcome

The dashboard closes the loop between a phone conversation and an actionable lead.

1 Lead appears after phone-number submission.
2 Call status is tracked through its lifecycle.
3 Project requirement and timeline are persisted.
4 Preferred language is retained.
5 Meeting interest drives qualification state.
6 Operator can inspect the result without replaying the call.

Field	Example
Project	Living room and kitchen renovation
Timeline	Two-three months later
Conversation	Hinglish
Status	Qualified
Designer meeting	Requested

Why the dashboard matters

The dashboard is backed by the same persistence path used by the live call. It demonstrates that the agent produces operational business data, not just a conversational demo.

Production-Oriented Engineering

The project was deliberately taken beyond a toy voice script.

Area	Implementation
Frontend	Next.js production build verified.
API	FastAPI lead, voice and dashboard routes.
Health	/health service health endpoint.
Configuration	Environment-based Exotel/public URL settings.
Webhooks	Separate save-lead and Exotel status endpoints.
Call states	Calling, ringing, in-progress, completed, failed, no-answer.
Error handling	Exotel errors can mark call failure.
Lead matching	Phone-based fallback when tool payload lacks phone.
Build validation	TypeScript compilation/static generation/routes verified.

Deployment boundary

The live demo currently uses local services plus ngrok as the public bridge. A permanent cloud deployment would replace ngrok with hosted HTTPS and add durable production persistence and observability.

Alternatives & Cost

The final stack was chosen to reduce integration and latency risk.

Alternative	Decision / reason
Browser-only voice	Rejected: does not satisfy real phone callback requirement.
Custom STT + LLM + TTS	Rejected: more orchestration and latency risk for this scope.
Generic voice stack	Rejected: Indian-language/code-mixed fit was weaker.
Manual follow-up	Rejected: fails immediate-response requirement.
Sarvam + Exotel	Selected: strong fit for Indian realtime voice + phone integration.

Cost

The assignment target was roughly █2–3/min, with █5–6/min acceptable if justified. The researched Sarvam Voice Agents pricing used during implementation was approximately █3.50/min, close to the target while reducing the need to maintain a custom realtime voice stack.

Cost philosophy

The goal is not minimum API cost at any price. A slightly higher platform cost can be justified when it reduces engineering complexity, latency risk, telephony work and operational burden.

The Journey: Idea → Live System

The project evolved from a simple assignment into a connected realtime workflow.

Stage	Evolution
1. Brief	Immediate human-like callback after phone submission.
2. Conversation design	60–90 sec flow, Shubh persona, natural close, multilingual behaviour.
3. Voice deployment	Sarvam agent configured and deployed.
4. Web integration	Next.js → FastAPI → outbound call.
5. Telephony	Exotel connected to Sarvam VoiceBot.
6. Extraction	save_lead_details added for structured business fields.
7. Dashboard	Operator-facing lead and qualification view.
8. Debugging	No-answer, webhook, ngrok and port-routing issues isolated.
9. Public path fix	ngrok changed to explicit 127.0.0.1:8000.
10. Verification	Local/public health, webhook, live call and persistence validated.

The biggest evolution was conceptual:

“Make an AI call” → “Make a complete lead-conversion loop.”

1 Customer intent
2 Immediate callback
3 Natural qualification
4 Structured lead
5 Designer-meeting intent
6 Operator visibility

Engineering Lessons

The strongest lessons came from real integration boundaries.

Lesson	Takeaway
Voice UX is a systems problem	Model quality cannot compensate for broken call setup, turn-taking or webhooks.
Keep the conversation narrow	A 60–90 sec business call should collect only decision-relevant information.
Structured outputs matter	They convert conversation into actionable CRM data.
Test public integrations separately	Local success does not prove external reachability.
Tunnel behaviour can look like an app bug	The ngrok warning initially obscured the routing problem.
Explicit network targets reduce ambiguity	127.0.0.1:8000 made the intended listener explicit.
Call status ≠ lead status	A completed call is not automatically a qualified lead.
Closing deserves explicit design	The final seconds determine whether the interaction feels natural.

Final System

The final implementation is a realtime lead-conversion pipeline.

1 Next.js visitor experience
2 FastAPI lead API
3 Exotel outbound telephony
4 Sarvam realtime voice agent
5 Structured extraction
6 FastAPI webhook persistence
7 Dashboard / operator visibility

Capability	State
Phone capture	Implemented
Immediate outbound call	Implemented
Natural Indian voice	Implemented
English / Hindi / Hinglish	Implemented
Interruptions	Enabled
Project + timeline capture	Implemented
Designer meeting intent	Implemented
Lead persistence	Implemented
Exotel callbacks	Implemented
Dashboard	Implemented
Public webhook	Validated through ngrok
Permanent cloud deployment	Next infrastructure step

FINAL TAKEAWAY

The strongest AI voice product is not a clever prompt alone. It is a reliable system around the prompt — one that can initiate the call, survive integration boundaries, capture useful information, and leave the business with a clear next action.

The page is the entry point. The call is the experience. The structured lead is the business outcome.

SUNRISE INTERIORS

From idea to live call.

A complete journey from assignment brief to realtime AI calling, telephony integration, structured lead capture and operator visibility.

PROJECT COMPLETE

Built, integrated, debugged, publicly validated, and prepared for final demonstration.

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